Consensus and leader election

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Sample chapter for the proposed course plan.

1 Learning outcome

Explain the role of a majority in a three-node consensus group.

Prerequisites: Lectures 05–06.

2 Core idea

Consensus lets participating nodes agree on decisions despite some failures. In Raft, a three-node group uses a majority of two. A node isolated from the other two cannot commit a new log entry by itself. See Ongaro and Ousterhout, Raft, sections 2 and 5.

3 Worked scenario

Three registration metadata nodes split into a connected pair and one isolated node. The pair can make progress once a suitable leader is established.

4 Discussion

Why should the isolated node avoid accepting independent committed updates?

5 Recap

Agreement and progress are separate properties; a partition can prevent progress for some nodes.